communications psychology

I did what? Testing whether self-attribution requires conscious awareness

John P. Veillette[1]*, Howard C. Nusbaum[1]

[1] Department of Psychology, The University of Chicago, Chicago, IL

* Corresponding author: John P. Veillette (johnv@uchicago.edu)

Abstract

The nature of self-awareness has been a topic of inquiry for thousands of years, with profound implications for law and ethics, as well as for understanding a host of neurological and psychiatric pathologies. An influential view in philosophy of mind is that the "self" is a construct of consciousness, its basic functions – such as the sense of agency, the capacity by which we attribute sensory events to our own control – cease when they fall out of awareness. An alternative view is that some core processes that constitute the self can operate outside of awareness, and self-awareness arises when these extant processes become contents of consciousness. We aim to test between these views empirically by investigating whether intentional binding – an implicit marker of sense of agency in which the perceived time of an action is shifted toward its sensory outcome – occurs even when the outcome is masked from conscious awareness.

Introduction

A prominent view amongst philosophers of mind is that the "self," as we experience it, is merely an illusion of our conscious minds [1]. While it is the case, in this view, that *all* of our conscious experiences are constructions insofar as we experience, for instance, unified perceptions of objects rather than the raw responses of our sensory receptors, the self is distinct from ordinary (non-hallucinatory) perceptions in that it does not correspond to anything that persists outside of conscious awareness. While the (non-)reality of selfhood is not a settled issue in philosophy [2,3], varying forms of this illusionist view have been endorsed [4], with some empirical evidence [5,6], in the broader cognitive sciences. However, we would argue this view is implicitly assumed more often than it is explicitly stated. For example, when Benjamin Libet famously presented evidence that the earliest neural activity associated with movement temporally preceded subjects' conscious awareness of intending to move [7], his findings were interpreted as a challenge to free will, as the conscious self seemed to be retroactively claiming credit for actions it did not initiate. While this interpretation of the Libet study has been thoroughly disputed on methodological grounds [8], the fundamental assumption that the unconscious process that initiated the action is clearly separable from the conscious self is rarely questioned [9]. That is, many scientists have tacitly ruled out the possible alternative that the core set of neural and cognitive processes that constitute that self may be able to operate independently of consciousness, and we become "self-aware" when those processes meet the same conditions that allow any other process or percept to appear in awareness [10,11] – thus, any sense of agency we consciously experience over an unconsciously initiated action must be illusory.

To be clear, what we are calling the "illusionist view" does not necessarily hold that *no* self-referential processes occur outside of awareness; the existence of implicit, unconscious self-monitoring processes – as for neuromuscular control or allostatic regulation – is typically acknowledged but differentiated from explicit processes that appear in and require consciousness [12]. Critically, the processes which are considered sufficient for "minimal self-awareness" – most often said to be the sense of agency (SoA), the experience of authoring actions and their consequences, and the sense of ownership (SoO)[13], the experience of having a body, though other ingredients have been argued to be necessary for explaining anomalous experiences of self [14] – must in some way require consciousness in the illusionist view. Conversely, the alternative we will call the "independence view" does not necessarily hold that *all* processes that make up the self can occur outside of conscious awareness, nor does it hold that the self is not constructed by a biophysical process. It merely posits that some of the processes that constitute the minimal self may not require consciousness to operate per se. This view extends from a sentiment in the wider natural sciences that a meaningful notion of self, particularly of *agency,* emerges from the same characteristics of biophysical organization that differentiate life, i.e. agentic material, from non-life; thus SoA is a legitimate awareness of an agentic capacity that exists at the scale of the organism rather than merely a confabulation of the conscious organism [15,16]. Most first-order theories of consciousness would be compatible with (but not imply) the independence view [10,11], as has been argued explicitly by some first-order theorists to justify how consciousness can be studied independently of self-awareness [17]. Other approaches to understanding consciousness, such as embodied or enactivist theories [18] and higher-order theories [19] of consciousness – some of which argue that the first-person perspective is what makes a percept conscious – may even require that a notion of self is a precondition for consciousness, rather than the other way around as in the illusionist view.

The critical distinction, then, is that for the illusionist, awareness is a precondition to *self,* not just to the *awareness of self*. Thus, were the illusionist to speculate how an organism first came to acquire self-awareness, they would say that the first self-aware organism must have first been aware and then subsequently developed SoA and SoO. In contrast, the independence view would allow for the possibility that some of these self-referential processes of which that organism became aware could have been inherited from an ancestor without conscious awareness. Becoming aware of these extant processes that generate the minimal self could "jump start" self-awareness. Even "high level" self-awareness that presumably does require consciousness may be influenced by these low-level processes, as evidenced by the finding that individual differences in the propensity to make positive SoA judgments in a sensorimotor task can predict a non-negligible amount of variance in subjects' narrative beliefs about one's own agency (although this relationship seems to be mediated by explicit metacognition) [20].

Until recently, findings in the SoA and SoO literatures have been largely compatible with the illusionist separation between unconscious, implicit self-monitoring and conscious, explicit self-monitoring. For instance, it is widely held that effective neuromuscular control requires an internal model of one's body to compensate for intrinsic delays in sensory systems, and that these internal models are updated by predicting the sensory feedback from one's own movement and cancelling out predicted sensations – such that non-suppressed sensations constitute prediction errors [21]. This comparator model accounts for the observation that, for instance, people cannot tickle themselves [22], and thus it can be said to discriminate between self- and other-caused sensations at a low level. As SoA is the phenomenological experience of a self-other distinction during action, this sort of predictive cancellation was one of the earliest proposed mechanisms for SoA [23,24], which would have conflicted with the illusionist view as this sort of sensory suppression based on sensorimotor contingency operates largely unconsciously and in essentially all organisms with a nervous system [25,26]. However, while neural markers of sensory suppression indeed *objectively* discriminate between self- and other-caused sensations [27], numerous studies have failed to find an association between these markers and *subjective* reports of agency [27–30]. This dissociation seems to support a division between unconscious and conscious self-monitoring processes, with only higher-level, abstract comparators and their prediction errors – which may not necessarily correspond to any obvious sensory suppression – informing SoA [24,31].

Recently, however, it has been demonstrated that when experimentally manipulating agency over a muscle movement using electrical stimulation (and opposed to over a downstream sensory consequence of movement as in typical paradigms), neural patterns indicative of pre-conscious sensory suppression are indeed predictive of subjective agency judgments [32], consistent with recent behavioural modelling results that suggest control detection processes may operate unconsciously [33]. Of course, a *pre*-conscious process – in which the subject eventually becomes conscious of the sensory stimulus and can report on it – is not the same as a fully *un*conscious process in which the subject will never become aware of the stimulus; it may be that this mechanism would never result in what we experience as SoA if not for some subsequent consciousness-dependent processing. However, this finding adds to a growing body of evidence that, while inconclusive, gives reason to doubt the illusionist account of self-awareness. For instance, it has been consistently reported that images of one's own face break through interocular suppression into awareness faster than other faces [34], as do arbitrary self-associated images [35] – although see a non-replication of the latter by Stein and colleagues [36] – suggesting that some process differentiates self- and other-related visual stimuli before they enter conscious awareness, potentially overlapping with the process that results in conscious SoO. Additionally, neural correlates of conscious own-face recognition [37–40] and own-name recognition [41] persist in the absence of conscious awareness of the visual stimulus. However, much like sensory suppression need not result in SoA, these self-other distinguishing neural signals may not result in SoO. Consider the zebra finch, in which a certain population of neurons is highly selective for the bird's own song, even when the bird is asleep or anesthetized [42]; without causal/behavioural evidence that the bird itself recognizes a song as their own when this population is responsive, it would be fallacious to conclude that the bird is self-aware based on neural data alone. Pfister and colleagues, in this vein, go on to demonstrate that own-name subliminal primes bias participant responses more than other-names in a speeded discrimination task [43], providing convincing evidence that these discriminations do have behavioural consequences but arguably falling short of demonstrating a self-ownership judgement (of the sort that, were we aware of it, we would experience as SoO) can be fully contained within an unconscious process. This is a natural limitation of studies of unconscious self-ownership discriminations, since it is particularly difficult to make claims about perceived ownership without taking self-report measures.

In contrast, the sense of agency is often measured not via self-report but via *intentional binding*, which can be measured if the subject is aware of either the action or the outcome – but not necessarily both. In a typical intentional binding paradigm, subjects make a button press, which is followed by a sensory stimulus – typically a tone but also a visual stimulus [44–47] – in an operant condition and not followed by anything in a baseline condition. They are then asked to report (e.g. by adjusting the hand of a clock that was spinning during the trial) the time the button press or the sensory consequence occurred. Haggard and colleagues were the first to report that the perceived time of an action was biased toward the time of its outcome and vice versa shortening the subjective interval between them [48]. Since then, this bias in temporal perception for intentional action has been shown to be a special case of a temporal binding that occurs between any two events perceived as a causal sequence [49]. The fact that this bias in the reported time of an action toward the time of its outcome interpretably tracks the perceived experience that a subject's action *caused* the outcome has made the intentional binding effect the gold-standard measurement of sense of agency literature, so much so that studies often do not solicit for self-report judgments at all [47,50,51].

Since the perceived time of the button press in an intentional binding paradigm (i.e. “action binding”) is usually measured independently of the time of the sensory consequence [52], this paradigm presents an intriguing opportunity: it is possible to measure temporal bias in the time of a button press, even if the subject is not consciously aware of the sensory consequence. To the extent that intentional binding is taken as a measure of SoA, this would suggest that subjects can implicitly attribute an action-contingent stimulus to their own action *even when they are unaware of the stimulus*. This would either provide evidence of that self-attributions of the sort we would consciously experience as SoA can occur outside of awareness, indicating that a core process underpinning minimal self-awareness does not itself require awareness, or it would require a reinterpretation of the highly influential intentional binding effect. To this end, we will ask subjects to complete a typical intentional binding paradigm in which they estimate the perceived time of their button press in both an operant and a baseline condition. In the first half of the experiment, however, we will mask the operant stimulus using continuous flash suppression (CFS), a form of interocular suppression (as in binocular rivalry) that allows stimuli presented to one eye to be robustly suppressed from awareness by presenting a dynamic stimulus to the opposite eye [53], though we do note the limitations of CFS ([54–56] and see **Methods: Design** for specific steps we have taken to account for these limitations) and the possibility of strongly reduced, rather than totally eliminated, awareness resulting from the manipulation. If subjects show evidence of action binding in conditions that suppress conscious awareness of the consequent stimulus (Hypothesis 1, see Table 1), this supports the view that intentional binding, and thus an implicit linking of action and outcome, does not require conscious awareness of the outcome. If the action binding effect measured in the unmasked condition is greater than that in the masked conditions (Hypothesis 2, see Table 1), which should occur in the event that Hypothesis 1 is false, then we will conclude that the intentional binding effect is facilitated by conscious awareness. Regardless, the present study stands to shed light on the illusive relationship between action-monitoring, self, and awareness in human beings.

It is worth elaborating on what exactly is under test here. Specifically, if Hypothesis 1 is indeed true, it does not imply the existence of an "unconscious SoA" that is ontologically equivalent to conscious SoA. (Whether the idea of a sense or feeling being unconscious even makes sense is, perhaps, a question best left to philosophers.) While intentional binding is frequently used as an implicit marker of SoA, it is not equivalent to the sense of agency itself; binding has been observed between cause-effect events in the absence of any intentional action [49]. In our case then, we would argue that an IB effect should be more conservatively interpreted as evidence that the brain has linked the action and the outcome, and under the independence view, one may experience SoA when they are conscious of this link. This finding would still corroborate a crucial prediction of the independence view, which does not hold that there is an "unconscious sense of agency" necessarily. Rather, it merely stipulates that SoA is a sense of *something* that exists regardless of whether one senses it, much like in ordinary sensory perception. For example, an experience of "pipe" normally refers to some *referent* that exists whether you are aware of it or not – namely, a physical object. This referent does not need to actually be a pipe for the perception to be non-hallucinatory; it could instead be a picture of a pipe, just as the referent for conscious SoA does not need to be an equivalent unconscious SoA. An unconscious causal link between action and outcome, suggested by an IB effect for an operant stimulus of which the subject is unaware, would constitute such a referent for agency. Such a finding would challenge illusionist perspectives which seem to take the lack of a plausible referent outside of consciousness as their premise [1,5], though some more nuanced illusionist perspectives may still be able to argue that the fact a *potential* referent is available does not mean it is actually referred to by conscious SoA. For example, Seth's (2021) account of self-awareness is rooted in a predictive coding account in which *all* perception is considered a "controlled hallucination" [4]. Such a possibility would be akin to a situation in which one failed to perceive an actual pipe, but nonetheless hallucinated an unrelated pipe, either by coincidence or because other sensory cues suggested an unobserved pipe was present. This is a particularly difficult argument to rebut empirically, and we do not think any possible outcome of the present experiment would do so conclusively. Nonetheless, results promise to place meaningful constraints on theories of self-awareness with respect to the contrast between illusionism and independence.

Methods

Ethics Information

All of the methods in the proposed study are in accordance with relevant safety and ethics guidelines for human subject research and have been approved by the Social and Behavioral Sciences Institutional Review Board at the University of Chicago (IRB23-1353). Informed consent will be obtained from each participant, and participants will be compensated for their participation with a credit that can be used to satisfy the research participation requirement of any course in Psychology offered at the university.

Pilot Data

We collected a small pilot sample consisting of $n = 5$ subjects to test our experimental procedures. Of the five subjects, all of them met the inclusion criteria outlined in **Sampling Plan**, which is to say that no subject reporting seeing significantly more masked stimuli on the operant-masked block than on the baseline-masked block, in which no masked stimuli were actually presented (Fisher's exact test, $p > 0.615$ for all subjects; see Table 2 for subject-by subject hit and false alarm rates). If the prevalence of subjects that would trigger our exclusion criteria (i.e. hit rate > false alarm rate with p < 0.05) in the sampled population is γ, then the number out of n subjects that trigger our exclusion criteria is distributed according to $\mathrm{Binomial}(n, 0.05(1-\gamma)+\gamma)$; with this likelihood and a uniform prior on prevalence γ, Bayes' rule dictates that the prevalence of subjects who would be excluded by the specified criteria is less than 39.3% with 95% posterior probability (estimated with the *bayesprev* Python package [57]). In other words, a majority of subjects are expected to survive our exclusion criteria with high probability. Even with this small number of subjects, we could already detect an intentional binding effect in the unmasked blocks (t(4) = 2.96, p = 0.027, median = 0.028 s); in fact, all five subjects individually showed a nominal intentional binding effect. Tests of Hypothesis 1 (t(4) = -1.51, p = 0.897, median = -0.011 s) and of Hypothesis 2 (t(4) = 2.04, p = 0.055, difference in medians = 0.039 s) were both non-significant at the significance levels we specify in Table 1, which is expected given the substantially higher sample size our power analyses (based on separate effect size measurements, see **Sampling Plan**) indicate are needed to detect (or estimate) either effect reliably. Rather, these pilot data were just to verify our task design works as intended. (Due to the very small pilot sample size, we report median effect sizes instead of means with confidence intervals, as we intended to report in a Stage 2 manuscript as described in **Analysis Plan**.)

Design

The experiment will consist of five within-subject blocks: (1) a calibration block, (2-3) operant-masked and baseline-masked blocks, in random order, and (4-5) operant-unmasked and a baseline-unmasked blocks, in the same order as Blocks 2-3. Randomization of the order in which subjects complete operant and baseline blocks ensures that intentional binding measurements are not systematically influenced by temporal drift in estimation bias.

During the experiment, subjects will wear red-blue anaglyph glasses for the presentation of stereoscopic visual stimuli, such that stimuli presented in red only appear to the left eye and blue stimuli only appear to the right eye. We will achieve suppression of the operant stimulus by presenting masks composed of many small squares, which update at a rate of 10 Hz, to the left eye, while the operant stimulus (a circle) is presented to the right eye for 200 milliseconds. Our intent is that the masks presented to the left eye totally suppress the competing image in awareness (see Figure 1a). However, any manipulation of interocular dominance – particularly those using anaglyph glasses, since red-blue light filtering is not perfect – must tune the contrast of the masked stimulus to ensure complete suppression, as "breakthrough" trials can occur during CFS [58]. Complete suppression can be verified by ensuring that they are sufficiently close to chance performance when asked what was presented to the suppressed eye [58,59]. Thus, in our calibration block, subjects will complete 100 trials of a discrimination task in which CFS masks are presented to the left eye for two seconds, during which a circle is presented to the right eye, in a random corner of the masked area, anytime between 0.25 and 1.75 seconds after CFS onset and remaining on screen for 200 ms. After each trial, subjects will be asked which side of the screen the blue circle appeared on. After each trial, we compute a Bayesian posterior for the log10-contrast threshold at which subjects can achieve 52.5% accuracy on the discrimination task using the QUEST algorithm with a Weibull psychometric function (with lapse rate = 0.01 and default slope = 3.5) [60]. Once the posterior is fit, we draw a random sample from the mean-truncated posterior distribution (see Figure 1b) and use it as the stimulus contrast for the next trial; this is a modified version of Thompson sampling [61], a state-of-the-art method for efficiently sampling when learning Bayesian posteriors. After 100 trials, we will use the 5^{th} percentile of the posterior distribution as the contrast for the remainder of the experiment, corresponding to a 95% posterior probability that discrimination accuracy at this threshold is below 52.5%. Assuming subjects get the answer right (100% accuracy) when they are aware of the stimulus and guess (50% accuracy), then this corresponds to a 95% probability that the rate of "breakthrough" trials in which subjects are aware of the masked stimulus is less than 5%.

(See **Sampling Plan** for consideration of why 5% is an acceptable breakthrough rate.) Notably, the use of an accuracy/discrimination threshold is more useful for our purposes than simply asking our subjects whether they see the circle stimulus, since it controls for the possibility that subjects may have an experience of the stimulus that falls above their threshold for minimal awareness but still below their certainty criterion for positively responding that they have seen it [59]. While using this addresses this potential confound during calibration, we should note that the additional controls we use in later blocks (i.e. asking subjects to report when they see a circle) do not account for this problem. We do not think there is any reason to believe that the nature of the main task (see below) would cause sensitivity thresholds to shift from where they were in the control block, but if this did occur, subjects may experience a quantitative (as opposed to a qualitative or complete) suppression of the operant stimulus from awareness. Minimally, CFS in this experiment serves to severely attenuate subjects' awareness of the stimulus to the point that they would not report that it is present when prompted.

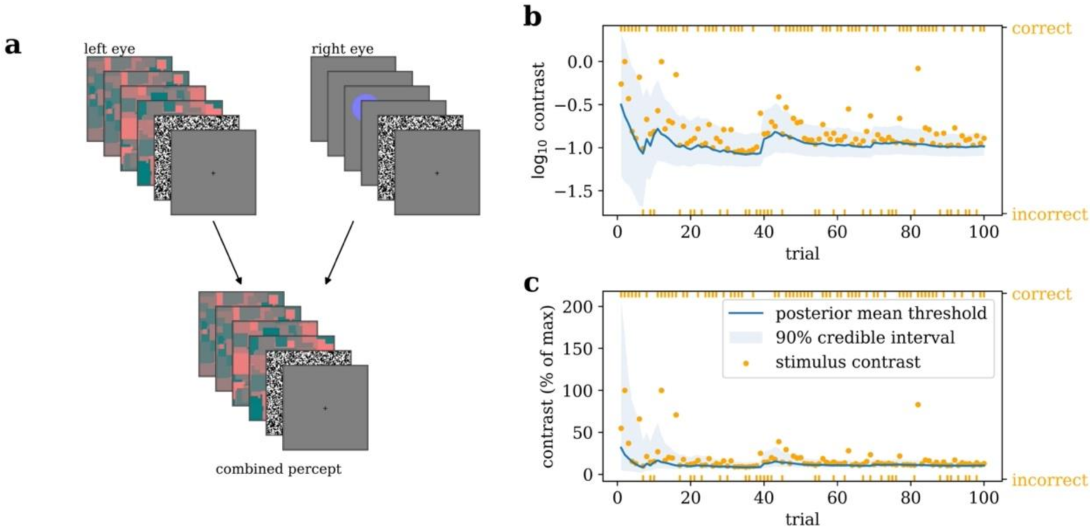

Figure 1. Calibration block. (a) In continuous flash suppression (CFS), high-contrast masks consisting of many squares varying in red pixel values are presented at 10 Hz to one eye, after which a single, static mask is displayed for 100 ms to reduce visual aftereffects. During CFS, a low-contrast blue circle is presented to the opposite eye, with the intent that it is suppressed from awareness when the visual system resolves the conflict between eyes. (b-c) During the calibration block, subjects report which side of the CFS mask they think the blue circle was presented on each trial. A Bayesian posterior is calculated for the stimulus contrast at which their discrimination accuracy is 52.5%, and the contrast for the next trial is drawn from the upper part of the posterior as in the example run illustrated here. At the end of the block, the 5th percentile of this posterior is used as the stimulus contrast for the remainder of the experiment, ensuring that subjects are aware of the stimulus on sufficiently few trials. The calibration curves shown here are from a single pilot subject.

Once we have determined the contrast for the main experiment, subjects will complete either the baseline-masked block (see Figure 2a) or the operant-masked block (Figure 2b), randomly selected, and then they will complete the other. In the baseline-masked block, CFS masks will be presented in the middle of an on-screen clock with a hand rotating around the outside. Any time following the first rotation of the clock, subjects will press the space bar on their keyboard, after which the CFS masks will end and the clock will stop rotating 1-2 seconds later, the clock hand will disappear for an additional second, and then it will reappear at a random location 45 to 60 degrees away from where it was when the key was pressed. Subjects will then be asked to adjust the clock hand back to where it was when they pressed the space bar. In the operant block only, a circle identical to that in the calibration block will appear in a corner of the masked area (which is randomly chosen on a per-subject basis but the same across all operant trials) will appear 150 ms after the subjects' key press or on the next screen refresh and persist for 200 ms. As an additional check to ensure subjects are not aware of the operant stimulus – in case, for instance, sensorimotor contingency alters the breakthrough threshold [26] – we will ask subjects after each masked trial whether they saw a blue circle, and we will randomly insert a few "decoy" trials in which a full-contrast circle appears on the first rotation of the clock. Each block will have 5 practice trials, 5 decoy trials, and 40 real trials (only real trials will be analysed).

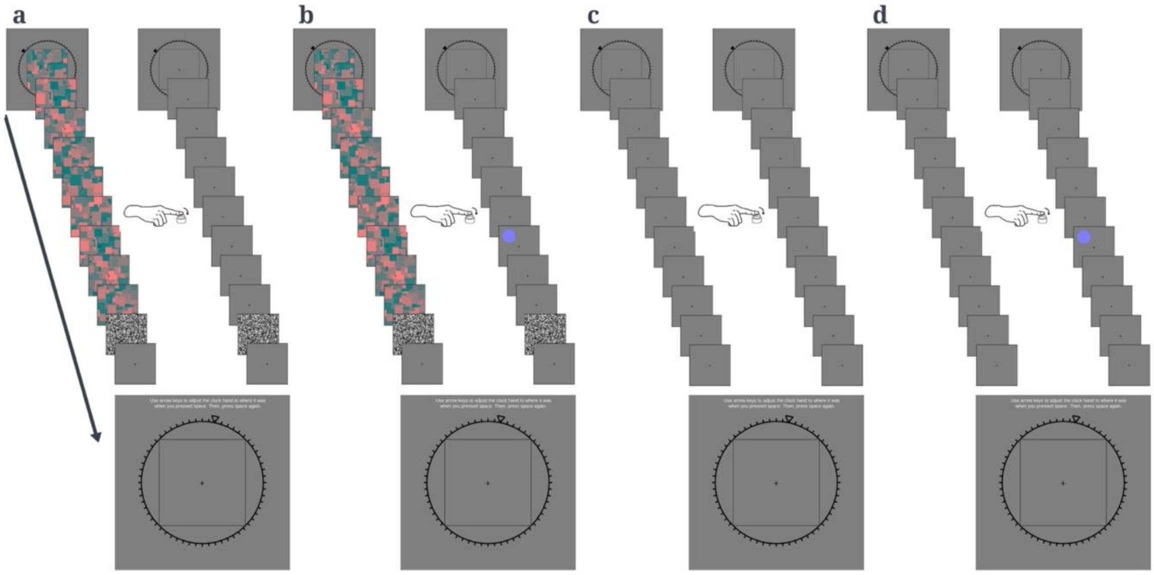

Figure 2. Intentional binding task. (a) In the baseline-masked block, CFS masks are presented to one eye in the middle of a rotating on-screen clock. The subject chooses when to press a button, 1-2 seconds after which CFS ends and the clock stops. The clock hand disappears for 1 second and reappears in a random location, after which subjects are prompted to adjust it back to its position when they pressed the button. (b) In the operant-masked condition, a circle is presented to the suppressed eye 150 ms after the subject makes the button press. (c-d) The baseline-unmasked and operant-unmasked conditions are identical to the masked versions, but without CFS so subjects are consciously aware of the operant stimulus.

After the masked blocks, subjects will complete baseline and operant blocks, consisting of 5 practice trials and 40 real trials (again, only real trials will be analysed), with no CFS masking (see Figures 2c-d). Baseline and operant blocks will be completed in the same order as the masked versions were completed so potential order effects are matched between masked and unmasked conditions. While no masks will be presented to the left eye in these blocks, the operant stimulus will still only be presented to the right eye and at the same contrast as in the masked blocks to ensure visual stimulation is otherwise identical. Thus, the unmasked blocks function as a positive control; estimation bias should differ between the operant-unmasked and baseline-unmasked blocks regardless of whether Hypothesis 1 or 2 is true, or we will have to conclude our paradigm was ineffective at eliciting the intentional binding effect.

The experimenter will be blind to the order in which the operant and baseline conditions are presented; randomization is handled entirely by the experiment code (see **Code Availability**), which fully implements the procedures described in this section.

Sampling Plan

We performed sample size calculations to ensure that tests of Hypothesis 1, Hypothesis 2, and our positive control (see **Analysis Plan**) all had statistical power of greater than 95%. Sample size calculations were performed using the *statsmodels* package in Python. In conjunction with these power analyses, we also conducted simulations to ensure that positive results for Hypothesis 1 are unlikely to be caused by "breakthrough" trials in which subjects were spuriously aware of the operant stimulus (though these simulations do not substitute for an in-task manipulation check, see **Design** and the last paragraph of this section).

In addition to the statistical Type 1 error inherent to null hypothesis significance tests, the possibility that the subject may become aware of the operant stimulus on some trials despite CFS masking introduces another source of false positives when testing Hypothesis 1. Thus, if we were to use the typical significance level of 0.05 for our statistical test, then the total false positive rate would necessarily be higher than 0.05. In this vein, we set the significance level for Hypothesis 1 to 0.01 to leave some cushion for error due to breakthrough trials. After we determined the sample size for the experiment, we conducted simulations to ensure the total false positive rate (Type I + awareness confound) would remain below 5%.

For Hypothesis 1, we performed power calculations using the meta-analytic effect size for action binding (i.e. difference in mean temporal estimation bias between operant and baseline conditions), Cohen's *d* = 0.451 [62]. For our statistical test (see **Analysis Plan)** to have 95% power at a significance level of 0.01, calculations showed that we would need 80 subjects.

For Hypothesis 2, we aim for 95% power *in the event that Hypothesis 1 is fals*e. That is, we want 95% power for concluding that the intentional binding effect is stronger in the unmasked conditions than in the masked conditions when the effect size is $d_{main} = 0.451$ (as above) in the unmasked conditions and $d_{null} = 0$ in the masked conditions. If we know the within-subject variance σ_{wi} and the between-subject variance σ_{bw} in the paired (operant – baseline) differences, then we can compute $d_{interaction} = d_{main} / \sqrt{1 + \sigma_{wi}^2 / \sigma_{bw}^2}$, which amounts to doubling the within-subject variance (since a difference-in-differences required four paired measurements instead of two). Since between-subject and within-subject variances are not available from the meta-analysis from which we drew the main effect size , we computed point estimates of these quantities from a 192 subject intentional binding dataset we have reported previously and made available on Open Science Framework [20]. We first computed the trial-by-trial within-subject variance as the mean of the variances of responses within each subjects' baseline block $\sigma_{trial} = 92.8$, then the per-block variance as $\sigma_{block} = \frac{\sigma_{trial}}{\sqrt{40}}$ by the central limit theorem. Finally, we computed within-subject variance for the paired differences (between two blocks) as $\sigma_{wi}^2 = 2\sigma_{block}^2$, and the between-subject variance as $\sigma_{bw}^2 = \sigma_{total}^2 - \sigma_{wi}^2 = (78.8)^2$. With these estimates in hand, we computed an effect size of $d_{interaction} = 0.44$, for which we would need 58 subjects to detect with 95% power at a significance level of 0.05.

For our positive control, the meta-analytic effect size is the same as for Hypothesis 1 but the significance level is set higher, at 0.05. Therefore, if we have 95% power to test Hypothesis 1, then we have greater than 95% power for our positive control. Since Hypothesis 1 requires more subjects to test than Hypothesis 2, we will use the sample size computed for Hypothesis 1 as our final sample size. Thus, we will collect n = 80 subjects with normal colour vision from the University of Chicago student community.

Lastly, we want to verify that a 5% rate of breakthrough trials (see **Design** for how we contain that rate) is not enough to unduly inflate our false positive rate for Hypothesis 1. To do this, we used σ_{bw} and σ_{trial} as calculated above to simulate two sets of 40 trials for 80 subjects. The simulated baseline-masked block had a mean estimation bias of zero, but timing estimates on each trial had standard deviation σ_{trial}. As we were simulating data under the null hypothesis, the simulated operant-masked block also had mean 0 and scale σ_{trial}, but a subject i would become "aware" of the operant stimulus on $k \sim$Binomial(40, 0.05) trials, which had mean $\mu_i \sim N(\mu, \sigma_{bw}^2)$, where $\mu = 25.7$ ms is the mean paired difference in estimation bias observed in our previous binding dataset [20]. We then test Hypothesis 1 on this simulated data as in **Analysis Plan**. Repeating this simulation procedure 10,000 times, we find that we falsely reject the null hypothesis 3.79% of the time, 1% of which is accounted for by the error rate of our statistical test, suggesting that breakthrough trials only inflate the false positive rate by about 2.79%.

As an additional quality check to ensure breakthrough trials do not confound our results (e.g. if threshold drops subsequent to initial measurement), we will compare the rate at which subjects report seeing a blue circle in the operant-masked block to that in which they report seeing a circle in the baseline-masked block, not including practice or decoy trials. Since the baseline block does not have an operant stimulus, its observed detection rate is a false alarm rate. If a subject's detection rate on the operant block exceeds their false alarm rate according to Fisher's exact test with a significance level of 0.05, we will conclude that they were aware of the operant stimulus on an unacceptable proportion of trials and exclude them from analysis. Such subjects will be replaced so that the usable sample size remains at *n* = 80.

Analysis Plan

For each subject, we will compute the Huber mean of the estimation bias (estimated event time minus true event time) within each block, excluding any trial in which subjects reported being aware of an operant stimulus. The Huber mean is an outlier-robust estimate of central tendency, eliminating the need to trim outlier trials from the data [63]. Paired differences Δ_{masked} between the operant-masked and baseline-masked conditions will be calculated, as will paired differences $\Delta_{unmasked}$ between the operant-unmasked and the baseline-unmasked conditions.

As a positive control, we will compare $\Delta_{unmasked}$ to the null hypothesis $H_0\colon \Delta_{unmasked} = 0$ using a one-tailed t-test at significance level 0.05. Then, to test Hypothesis 1, we will compare Δ_{masked} to the null hypothesis $H_0\colon \Delta_{masked} = 0$ using a one-tailed t-test at significance level 0.01 (see **Sampling Plan** for explanation of reduced significance level).

To test Hypothesis 2, i.e. that $\Delta_{unmasked}$ is greater than Δ_{masked}, we will compute the paired difference-in-differences $\Delta_\Delta = \Delta_{unmasked} - \Delta_{masked}$, and we will again compare this quantity to chance using a one-tailed t-test.

We will then generate bootstrapped distributions (5,000 samples) for $\Delta_{unmasked}$, Δ_{masked}, and Δ_Δ from which we will obtain 95% confidence intervals for each quantity, as well as Cohen's *d* estimates and confidence intervals for $\Delta_{unmasked}$ and Δ_{masked}, using the DABEST package for estimation statistics. (DABEST does not currently implement Cohen's *d* for differences-in-differences.) Confidence intervals will be visualized as are the hypothetical results shown in Figure 3.

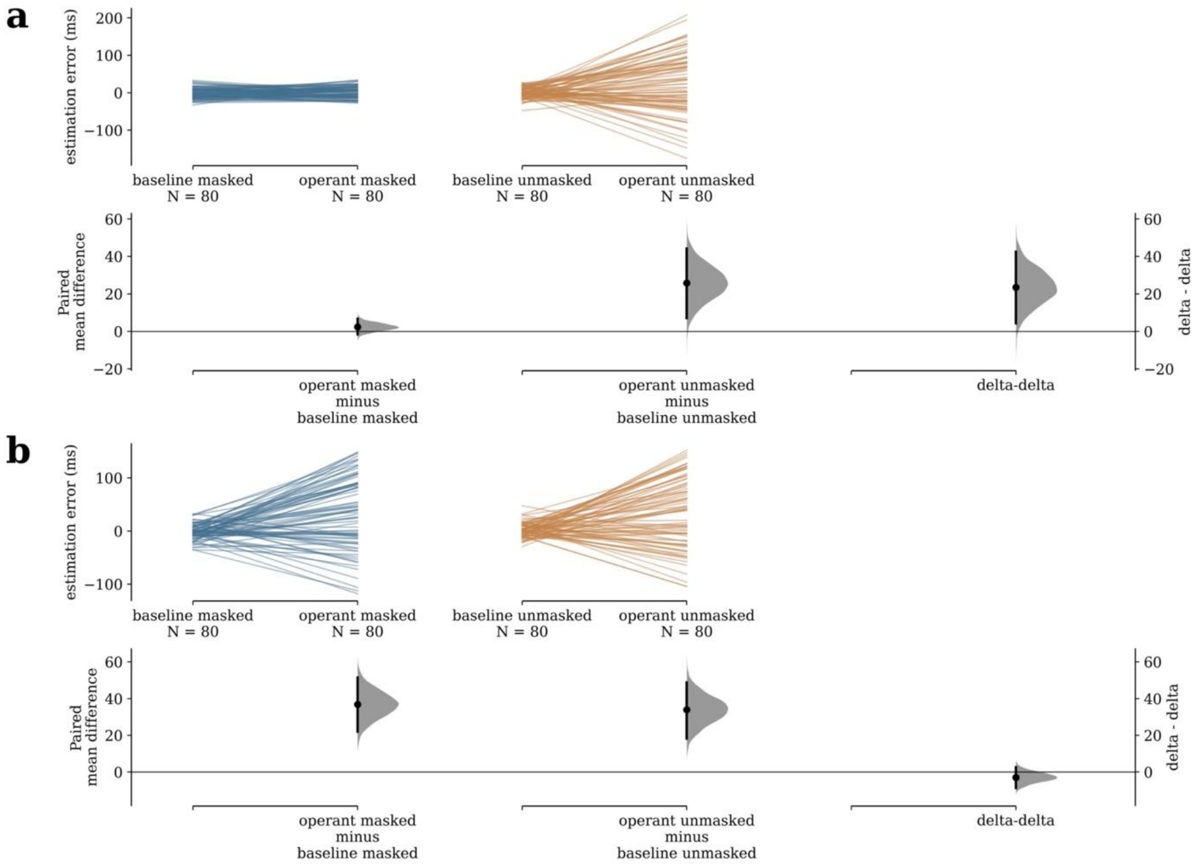

Figure 3. Hypothetical outcomes (simulated data). (a) If intentional binding requires conscious awareness of the operant stimulus, i.e. Hypothesis 1 is false, then estimation bias will be the same in both masked blocks, but will be positively biased in the operant-unmasked block. The difference-in-differences, then, will be positive as per Hypothesis 2. (b) Alternatively, if Hypothesis 1 is true, evidence of binding will be found in both blocks, and the difference-in-differences is unlikely to be very different from zero, i.e. Hypothesis 2 is false. The experiment is designed such that, if one hypothesis is strictly false (i.e. zero effect size), we should be well-powered to detect the competing hypothesis.

Data availability

All raw data will be shared in a public repository upon submission of the Stage 2 manuscript. Pilot data is already included in the same repository as the analysis code (https://zenodo.org/doi/10.5281/zenodo.13273592).

Code availability

The full code for conducting the experiment is available on *Zenodo* at https://doi.org/10.5281/zenodo.13273591, and code for sample size calculations, data simulation, and analysis of simulated and pilot data – which is the same analysis to be applied to the final dataset once collected – is available at https://zenodo.org/doi/10.5281/zenodo.13273592. Versions of both repositories at the time of preregistration are permanently archived with unique DOIs, as the final versions will be upon Stage II submission.

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Author contributions

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Competing interests

The authors declare no competing interests.

Table 1. Design Table

Question	Hypothesis	Sampling plan (e.g. power analysis)	Analysis Plan	Interpretation given to different outcomes
Does intentional binding require conscious awareness?	**Hypothesis 1** Estimation bias will be greater on operant-masked trials than on baseline-masked trials.	95% power for Cohen's *d* = 0.45 is obtained with 80 subjects, using a significance level of 0.01. Subjects will be excluded (and replaced) if they report seeing more circles on operant-masked trials than on baseline-masked trials (Fisher's exact test, sig. level 0.05).	Mean estimation bias (estimated time minus true event time) will be computed for each block, and a one-sided *t*-test will be used to compare the paired differences Δ_{masked} between operant and baseline conditions to $H_0\colon \Delta_{masked} = 0$.	If $\Delta_{masked} > 0$ at $p < 0.01$, we will conclude that intentional binding can occur in the absence of conscious awareness. If $\Delta_{masked} > 0$ at $p > 0.01$ or $\Delta_{masked} < 0$, we will not interpret the null result.
Does conscious awareness facilitate intentional binding?	**Hypothesis 2** The magnitude of the intentional binding effect will be greater in the unmasked conditions than it is in the masked conditions.	95% power for Cohen's *d* = 0.44 is obtained with 58 subjects, using a significance level of 0.05, but we collect 80 subjects to ensure sufficient power for Hypothesis 1.	The paired difference in differences $\Delta_\Delta = \Delta_{unmasked} - \Delta_{masked}$ will be compared to $H_0\colon \Delta_\Delta = 0$ using a one-sided *t*-test.	If $\Delta_\Delta > 0$ with $p < 0.05$, we will conclude that conscious awareness facilitates the intentional binding effect. If $\Delta_\Delta > 0$ with $p > 0.05$ or $\Delta_\Delta < 0$, we will not interpret the null result.

Table 2. Per-subject Detection Rates in Pilot Sample.

Hit rate on catch trials	Hit rate on operant trials (HR)	False alarm rate on baseline trials (FA)	Fisher's exact test *p*-value (H_0: HR = FA)
0.900	0.075	0.075	1.000
0.700	0.000	0.000	1.000
1.000	0.075	0.025	0.615
1.000	0.000	0.025	1.000
1.000	0.025	0.075	0.615